

Question #01

orthogonalize and unitization with the schmidt method.

$$x_1 = \begin{bmatrix} 1 \\ i \\ 0 \end{bmatrix}, x_2 = \begin{bmatrix} 1 \\ 0 \\ i \end{bmatrix}, x_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$y_1 = x_1 = \begin{bmatrix} 1 \\ i \\ 0 \end{bmatrix}$$

$$y_2 = x_2 - \frac{[x_2 y_1]}{[y_1 y_1]} y_1$$

$$[x_2 y_1] = y_1^H x_2$$

$$= \begin{bmatrix} 1 & -i & 0 \end{bmatrix} \begin{bmatrix} 1 \\ i \\ 0 \end{bmatrix}$$

$$= 1 + 0 + 0 = 1$$

$$[y_1 y_1] = y_1^H y_1$$

$$= \begin{bmatrix} 1 & -i & 0 \end{bmatrix} \begin{bmatrix} 1 \\ i \\ 0 \end{bmatrix}$$

$$= 1 - i^2 + 0$$

$$\Rightarrow 1 - (-1) = 2$$

$$y_2 = \begin{bmatrix} 1 \\ 0 \\ i \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 \\ i \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 0 \\ i \end{bmatrix} - \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2}i \\ 0 \end{bmatrix}$$

$$y_2 = \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2}i \\ i \end{bmatrix}$$

$$[x_3 y_1] = y_1^H x_3$$

$$= \begin{bmatrix} 1 & -i & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = 0$$

$$y_3 = x_3 - \frac{[x_3 y_1]}{[y_1 y_1]} y_1 - \frac{[x_3 y_2]}{[y_2 y_2]} y_2$$

$$[x_3 y_2] = y_2^H x_3$$

$$= \begin{bmatrix} \frac{1}{2} & \frac{1}{2}i & -i \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$= 0 + 0 - i \Rightarrow -i$$

$$[y_2 y_2] = \begin{bmatrix} \frac{1}{2} & \frac{1}{2}i & -i \end{bmatrix} \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2}i \\ i \end{bmatrix}$$

$$= \frac{1}{4} - \frac{1}{4}i^2 - i^2 = \frac{1}{4} + \frac{1}{4} + 1 = \frac{3}{2}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} - \frac{0}{2} \begin{bmatrix} 1 \\ i \\ 0 \end{bmatrix} - \frac{-i}{\frac{3}{2}} \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2}i \\ i \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} - 0 + \frac{2}{3}i \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2}i \\ i \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} + \begin{bmatrix} \frac{1}{3}i \\ \frac{1}{3} \\ -\frac{2}{3} \end{bmatrix} \Rightarrow \begin{bmatrix} \frac{1}{3}i \\ \frac{1}{3} \\ \frac{1}{3} \end{bmatrix}$$

we can also prove that these are orthogonal to each other.
for example if $y_1 \perp y_2$ then $[y_1, y_2]$ should be zero.

$$\begin{aligned} [y_1, y_2] &= y_2^H \cdot y_1 = \begin{bmatrix} \frac{1}{2} & \frac{1}{2}i & -i \end{bmatrix} \cdot \begin{bmatrix} 1 \\ i \\ 0 \end{bmatrix} \\ &= \frac{1}{2} + \frac{1}{2}i^2 - 0 \\ &= \frac{1}{2} - \frac{1}{2} = 0 \end{aligned}$$

Now $z_1 = \frac{y_1}{\|y_1\|_2} = \frac{\begin{bmatrix} 1 \\ i \\ 0 \end{bmatrix}}{\sqrt{[y_1, y_1]}}$

$$= \frac{\begin{bmatrix} 1 \\ i \\ 0 \end{bmatrix}}{\sqrt{2}} \Rightarrow \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ i \\ 0 \end{bmatrix}$$

$$z_2 = \frac{y_2}{\|y_2\|_2} = \frac{\begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2}i \\ i \end{bmatrix}}{\sqrt{\frac{3}{2}}} = \sqrt{\frac{2}{3}} \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2}i \\ i \end{bmatrix}$$

we can further simplify

$$= \frac{\sqrt{2}}{\sqrt{3}} \cdot \frac{1}{2} \begin{bmatrix} 1 \\ -i \\ 2i \end{bmatrix} = \frac{\sqrt{2}}{\sqrt{3}} \cdot \frac{1}{\sqrt{2} \cdot \sqrt{2}} \begin{bmatrix} 1 \\ -i \\ 2i \end{bmatrix}$$

$$z_2 = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ -i \\ 2i \end{bmatrix}$$

$$z_3 = \frac{y_3}{\|y_3\|_2} = \frac{\begin{bmatrix} \frac{i}{3} \\ \frac{1}{3} \\ -\frac{1}{3} \end{bmatrix}}{\sqrt{\frac{1}{3}}}$$

$$[y_3 y_3] = y_3^H y_3$$

$$= \begin{bmatrix} -\frac{i}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix} \begin{bmatrix} \frac{i}{3} \\ \frac{1}{3} \\ -\frac{1}{3} \end{bmatrix}$$

$$= \frac{-i^2}{9} + \frac{1}{9} + \frac{1}{9} = \frac{1}{3}$$

$$= \frac{\sqrt{3}}{3} \begin{bmatrix} i \\ 1 \\ 1 \end{bmatrix}$$

we can also check this.
(unitization)

$$[z_1 z_1] = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -i & 0 \end{bmatrix} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ i \\ 0 \end{bmatrix}$$

$$= \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} (1 - i^2 + 0) = \frac{1}{2} \cdot 2 = 1$$

Question # (same method
Different matrix)

$$x_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, x_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, x_3 = \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix}, x_4 = \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \end{bmatrix}$$

we know that

$$y_1 = x_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

for

$$y_2 = x_2 - \frac{[x_2 y_1]}{[y_1 y_1]} y_1$$

(in previous Question "i" so in y_i^H we changed signs of "i" values.

$$y_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} - \begin{bmatrix} \frac{1}{2} \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$y_2 = \begin{bmatrix} \frac{1}{2} \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

$$[x_2 y_1] = y_1^H x_2$$

$$= \begin{bmatrix} 1 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

$$= 1 + 0 + 0 + 0 = 1$$

$$[y_1 y_1] = y_1^H y_1$$

$$= \begin{bmatrix} 1 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$= 2$$

Now

$$y_3 = x_3 - \frac{[x_3 y_1]}{[y_1 y_1]} y_1 - \frac{[x_3 y_2]}{[y_2 y_2]} y_2$$

$$[x_3 y_1] = y_1^H x_3$$

$$= \begin{bmatrix} 1 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$$-1 + 0 + 0 + 0 = -1$$

$$y_3 = \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix} - \frac{(-1)}{2} \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} - \frac{(\frac{1}{2})}{\frac{3}{2}} \begin{bmatrix} \frac{1}{2} \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + \frac{1}{3} \begin{bmatrix} \frac{1}{2} \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix} + \begin{bmatrix} \frac{1}{2} \\ 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} \frac{1}{6} \\ 0 \\ \frac{1}{3} \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} -\frac{1}{3} \\ 0 \\ \frac{1}{3} \\ 1 \end{bmatrix}$$

$$[x_3 y_2] = y_2^H x_3$$

$$= \begin{bmatrix} \frac{1}{2} & 0 & \frac{1}{2} & 0 \end{bmatrix} \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$$= -\frac{1}{2} + 0 + 0 + 0 = -\frac{1}{2}$$

$$[y_2 y_2] = y_2^H y_2$$

$$= \begin{bmatrix} \frac{1}{2} & 0 & \frac{1}{2} & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{2} \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

$$= \frac{1}{4} + 0 + \frac{1}{4} + 0$$

$$= \frac{2}{4} = \frac{1}{2}$$

$$y_4 = x_4 - \frac{[x_4 y_1]}{[y_1 y_1]} y_1 - \frac{[x_4 y_2]}{[y_2 y_2]} y_2 - \frac{[x_4 y_3]}{[y_3 y_3]} y_3$$

$$[x_4 y_1] = \begin{bmatrix} 1 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \end{bmatrix} = 1 - 1 = 0$$

$$[x_4 y_2] = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \end{bmatrix} = \frac{1}{2} + \frac{1}{2} - 1 + 0 = 0$$

$$[x_4 y_3] = \begin{bmatrix} -\frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \end{bmatrix} = -\frac{1}{3} - \frac{1}{3} - \frac{1}{3} + 1 = 0$$

So

$$y_4 = \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \end{bmatrix} - 0 - 0 - 0 = \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \end{bmatrix}$$

Calculate

$$z_1 = \frac{y_1}{\|y_1\|_2}$$

$$z_2 = \frac{y_2}{\|y_2\|_2}$$

$$z_3 = \frac{y_3}{\|y_3\|_2}$$

and check

$$y_1 \perp y_2$$

Same method.

$$[z_1 z_1]$$



- to live, to stay
- * $\leq$ The Column Sum which give maximum answer.
- * $\leq$ The row Sum which give maximum answer.

Question #02

$$\text{let } A = \begin{bmatrix} 2i & 0 & -3 \\ 5 & 3 & 2 \\ -1 & 2 & 2 \end{bmatrix}$$

calculate $m_1, F, m_\infty, =1, \infty$

$$\textcircled{1} \|A\|_{m_1} = \sum_{i=1}^3 \sum_{j=1}^3 |a_{ij}| \quad \left(\begin{array}{l} \text{Sum of all} \\ \text{the absolute values} \\ \text{of elements.} \end{array} \right)$$

$$= |2i| + |0| + |-3| + |5| + |3| + |2| + |-1| + |2| + |2|$$

$$= 20$$

$$\textcircled{2} \|A\|_F = \sqrt{\sum_{i=1}^3 \sum_{j=1}^3 |a_{ij}|^2} \quad \left(\begin{array}{l} \text{sum of square} \\ \text{of all values} \end{array} \right)$$

$$= \sqrt{|2i|^2 + |0|^2 + |-3|^2 + |5|^2 + |3|^2 + |2|^2 + |-1|^2 + |2|^2 + |2|^2}$$

$$= \sqrt{60}$$

$$\textcircled{3} \|A\|_{m_\infty} = \max_{i,j} \{m, n\} \cdot \max_{i,j} |a_{ij}|$$

$$= 3 \cdot 5$$

$$= 15$$

$\max \{m, n\}$
 $\downarrow$
 maximum
 order of
 row or
 column

$$\textcircled{4} \|A\|_1 = \max_j \sum_{i=1}^3 |a_{ij}| = |2i| + |5| + |-1| = 8$$

column

3

$$\textcircled{5} \|A\|_\infty = \max_i \sum_{j=1}^3 |a_{ij}| = |5| + |3| + |2|$$

$$= 10$$

$\max_{i,j}$

maximum number
in rows or columns

5

Question#04 The condition number of a matrix.

$$A = \begin{bmatrix} 7 & 5 \\ 5 & 9 \end{bmatrix}$$

Calculate $\text{cond}_1(A)$, $\text{cond}_\infty(A)$, $\text{cond}_2(A)$

① $\text{Cond}_1(A) = \|A\|_1 \cdot \|A^{-1}\|_1$

$$A^{-1} = \frac{\text{adj} A}{\det A} = \frac{\begin{bmatrix} 9 & -5 \\ -5 & 7 \end{bmatrix}}{63-25} = \begin{bmatrix} \frac{9}{38} & -\frac{5}{38} \\ -\frac{5}{38} & \frac{7}{38} \end{bmatrix}$$

Compute

$$\|A\|_1$$

$$\text{Column 1 Sum} = |7| + |5| = 12$$

$$\text{Column 2 Sum} = |5| + |9| = 14$$

$$\text{So } \|A\|_1 = 14$$

Compute

$$\|A^{-1}\|_1$$

$$\text{Column 1 Sum} = \left| \frac{9}{38} \right| + \left| -\frac{5}{38} \right| = \frac{14}{38}$$

$$\text{Column 2 Sum} = \left| -\frac{5}{38} \right| + \left| \frac{7}{38} \right| = \frac{12}{38}$$

$$\text{So } \|A^{-1}\|_1 = \frac{14}{38} = 0.3684$$

Now

$$\text{Cond}_1(A) = \|A\|_1 \cdot \|A^{-1}\|_1$$

$$= 14 \cdot 0.3684$$

$$= 5.1579$$

2 $\text{Cond}_\infty(A)$

$$\text{Cond}_\infty(A) = \|A\|_\infty \cdot \|A^{-1}\|_\infty$$

$$A^{-1} = \begin{bmatrix} \frac{9}{38} & -\frac{5}{38} \\ -\frac{5}{38} & \frac{1}{38} \end{bmatrix}$$

Compute

$$\|A\|_\infty = \max \left(\sum_{j=1}^n |a_{ij}| \right)$$

$$\text{Row 1 Sum} = 12$$

$$\text{Row 2 Sum} = 14$$

$$\text{So } \|A\|_\infty = 14$$

compute $\|A^{-1}\|_\infty$

$$\text{Row 1 Sum} = \frac{14}{38}$$

$$\text{Row 2 Sum} = \frac{12}{38}$$

$$\text{So } \|A^{-1}\|_\infty = \frac{14}{38} = 0.3684$$

Now

$$\text{Cond}_\infty(A) = \|A\|_\infty \cdot \|A^{-1}\|_\infty$$

$$= 14 \cdot 0.3684$$

$$= 5.1579$$

3 $\text{Cond}_2(A)$

if $A^T = A$ and $\lambda_1 > \lambda_2 > 0$
then

$$\text{Cond}_2(A) = \frac{\lambda_1}{\lambda_2}$$

$$\text{as } A = \begin{bmatrix} 7 & 5 \\ 5 & 9 \end{bmatrix} ; A^T = A$$

So,

Compute $|hI - A| = 0$

$$hI = \begin{bmatrix} h & 0 \\ 0 & h \end{bmatrix}$$

$$hI - A = \begin{bmatrix} h & 0 \\ 0 & h \end{bmatrix} - \begin{bmatrix} 7 & 5 \\ 5 & 9 \end{bmatrix}$$

$$|hI - A| = \begin{vmatrix} h-7 & -5 \\ -5 & h-9 \end{vmatrix} \Rightarrow h^2 - 9h - 7h + 63 - 25$$

$$= h^2 - 16h + 38$$

$$a = 1, b = -16, c = 38$$

Using quadratic formula.

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-(-16) \pm \sqrt{(-16)^2 - 4(1)(38)}}{2(1)}$$

$$= \frac{16 \pm \sqrt{104}}{2} \Rightarrow \frac{16 \pm 2\sqrt{26}}{2} = 8 \pm \sqrt{26}$$

$$\lambda_1 = 8 + \sqrt{26}, \lambda_2 = 8 - \sqrt{26}$$

$$\text{Cond}_2(A) = \frac{\sigma_{\max}}{\sigma_{\min}}$$

For symmetric matrix ($A^T = A$)

$$\text{Cond}_2(A) = \frac{8 + \sqrt{26}}{8 - \sqrt{26}}$$

$$\sigma_{\max} = \lambda_1$$

$$\sigma_{\min} = \lambda_2$$

★ one more important Example

$$A = \begin{bmatrix} 2 & 4 \\ 4 & 7 \end{bmatrix} \text{ compute}$$

Hint.

$$A^T = A$$

Question #05

Compute e^A , e^{AT} , $\cos At$, $\sin At$

$$A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

$$\psi(\lambda) = |\lambda I - A| = 0$$

$$|\lambda I - A| = \begin{vmatrix} \lambda & -1 \\ 1 & \lambda \end{vmatrix} \Rightarrow \lambda^2 - (-1)$$

$$\psi(\lambda) = \lambda^2 + 1 = 0$$

$$\psi(A) = A^2 + I = 0$$

$$A^2 + I = 0 \Rightarrow A^2 = -I$$

$$A = A$$

$$A^5 = A^2 \cdot A^3$$

$$A^2 = -I$$

$$= (-I)(-A) = A$$

$$A^3 = A^2 \cdot A = (-I) \cdot A = -A$$

$$A^6 = A^2 \cdot A^2 \cdot A^2$$

$$A^4 = A^2 \cdot A^2 = (-I)(-I) = I$$

$$= (-I)(-I)(-I) = -I$$

$$A^7 = -A, A^8 = I$$

Now. By Taylor Series Expansion.

$$(1) e^A = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \frac{A^4}{4!} + \dots$$

$$(2) e^{AT} = I + At + \frac{A^2 t^2}{2!} + \frac{A^3 t^3}{3!} + \frac{A^4 t^4}{4!} + \dots$$

$$(3) \sin At = At - \frac{A^3 t^3}{3!} + \frac{A^5 t^5}{5!} - \frac{A^7 t^7}{7!} + \dots$$

$$(4) \cos At = I - \frac{A^2 t^2}{2!} + \frac{A^4 t^4}{4!} - \frac{A^6 t^6}{6!} + \dots$$

Remember these.

$$1 \quad e^A = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \frac{A^4}{4!} + \dots$$

$$= I + A + \frac{(-I)}{2!} + \frac{(-A)}{3!} + \frac{I}{4!} + \frac{A}{5!} + \dots$$

$$e^A = I \left(1 - \frac{1}{2!} + \frac{1}{4!} - \dots \right) + A \left(\frac{1}{3!} - \frac{1}{5!} + \dots \right)$$

$\downarrow$ $\cos 1$
 $\downarrow$ $\sin 1$

$$e^A = I \cos 1 + A \sin 1$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \cos 1 + \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \sin 1 = \begin{bmatrix} \cos 1 & \sin 1 \\ -\sin 1 & \cos 1 \end{bmatrix}$$

$$\underline{2} \quad e^{At} = I + At + \frac{A^2 t^2}{2!} + \frac{A^3 t^3}{3!} + \frac{A^4 t^4}{4!} + \dots$$

$$= I + At + \frac{(-I)t^2}{2!} + \frac{(-A)t^3}{3!} + \frac{I t^4}{4!} + \dots$$

$$= I \left(1 - \frac{t^2}{2!} + \frac{t^4}{4!} - \dots \right) + A \left(t - \frac{t^3}{3!} + \frac{t^5}{5!} - \dots \right)$$

$$= I \cdot \cos t + A \cdot \sin t$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \cos t + \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \sin t$$

$$= \begin{bmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{bmatrix}$$

$$\underline{\underline{3}} \quad \sin At = At - \frac{A^3 t^3}{3!} + \frac{A^5 t^5}{5!} - \frac{A^7 t^7}{7!} + \dots$$

$$= At + \frac{(A)t^3}{3!} + \frac{(A)t^5}{5!} + \frac{(A)t^7}{7!} + \dots$$

$$= A \left(t + \frac{t^3}{3!} + \frac{t^5}{5!} + \frac{t^7}{7!} + \dots \right)$$

↓

formula of sinht

So $\sin At = A \sin ht$

$$= \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \sin ht = \begin{bmatrix} 0 & \sin ht \\ -\sin ht & \cos ht \end{bmatrix}$$

$$\underline{\underline{4}} \quad \cos At$$

$$\cos At = I - \frac{A^2 t^2}{2!} + \frac{A^4 t^4}{4!} - \frac{A^6 t^6}{6!} + \dots$$

$$\cos At = I - \frac{(-I)t^2}{2!} + \frac{(I)t^4}{4!} + \dots$$

$$\cos At = I \left(1 + \frac{t^2}{2!} + \frac{t^4}{4!} + \dots \right)$$

$$= I \cdot \cos ht$$

$$\cos At = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \cos ht = \begin{bmatrix} \cos ht & 0 \\ 0 & \cos ht \end{bmatrix}$$

$$\cos A = \begin{bmatrix} \cos 1 & 0 \\ 0 & \cos 1 \end{bmatrix}$$

Question #06

Let $e^{At} = \begin{bmatrix} 2e^{2t} - e^t & e^{2t} - e^t & e^t - e^{2t} \\ e^{2t} - e^t & 2e^{2t} - e^t & e^t - e^{2t} \\ 3e^{2t} - 3e^t & 3e^{2t} - 3e^t & 3e^t - 2e^{2t} \end{bmatrix}$

find matrix A.

$$A = (e^{At})' \Big|_{t=0}$$

1 Differentiate e^{At} w.r.t t .

2 Evaluate at $t=0$

Proof $A = (e^{At})' \Big|_{t=0}$

$$= \frac{d}{dt}(e^{At}) \Rightarrow e^{At} \frac{d}{dt}(At)$$

$$= e^{At} \cdot A(1) \Rightarrow Ae^{At} \Big|_{t=0} = Ae^{A(0)} = A \cdot I = A$$

So

$$A = \frac{d}{dt}(e^{At}) \Big|_{t=0}$$

$$\frac{d}{dt} \begin{bmatrix} 2e^{2t} - e^t & e^{2t} - e^t & e^t - e^{2t} \\ e^{2t} - e^t & 2e^{2t} - e^t & e^t - e^{2t} \\ 3e^{2t} - 3e^t & 3e^{2t} - 3e^t & 3e^t - 2e^{2t} \end{bmatrix}$$

$$= \begin{bmatrix} 4e^{2t} - e^t & 2e^{2t} - e^t & e^t - 4e^{2t} \\ 2e^{2t} - e^t & 4e^{2t} - e^t & e^t - 2e^{2t} \\ 6e^{2t} - 3e^t & 6e^{2t} - 3e^t & 3e^t - 4e^{2t} \end{bmatrix}$$

$$t=0$$

$$= \begin{bmatrix} 4-1 & 2-1 & 1-2 \\ 2-1 & 4-1 & 1-2 \\ 6-3 & 6-3 & 3-4 \end{bmatrix} = \begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & -1 \\ 3 & 3 & -1 \end{bmatrix}$$

Question #08

Matrix Kronecker

$$\text{Let } A = \begin{bmatrix} 1 & 3 & 3 \\ 3 & 1 & 3 \\ 3 & 3 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

calculate eigen values of $A \otimes B$.

[Hint: calculate eigen values of both and then multiply]

$$\psi(\lambda) = |\lambda I - A| = 0$$

$$\lambda I = \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$\lambda I - A = \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} - \begin{bmatrix} 1 & 3 & 3 \\ 3 & 1 & 3 \\ 3 & 3 & 1 \end{bmatrix}$$

$$|\lambda I - A| = \begin{vmatrix} \lambda - 1 & -3 & -3 \\ -3 & \lambda - 1 & -3 \\ -3 & -3 & \lambda - 1 \end{vmatrix}$$

Expanding by R_1

$$= (\lambda - 1) \begin{vmatrix} \lambda - 1 & -3 \\ -3 & \lambda - 1 \end{vmatrix} - (-3) \begin{vmatrix} -3 & -3 \\ -3 & \lambda - 1 \end{vmatrix} + (-3) \begin{vmatrix} -3 & \lambda - 1 \\ -3 & -3 \end{vmatrix}$$

$$= (\lambda - 1) (\lambda^2 - 2\lambda + 1 - 9) + 3(-3\lambda + 3 - 9) - 3(9 + 3\lambda - 3)$$

$$= (\lambda - 1) (\lambda^2 - 2\lambda - 8) - 9\lambda - 18 - 18 - 9\lambda$$

$$= (\lambda - 1) (\lambda - 4) (\lambda + 2) - 18\lambda - 36$$

$$= (\lambda - 1) (\lambda - 4) (\lambda + 2) - 18(\lambda + 2)$$

$$= (\lambda + 2) [(\lambda - 4) (\lambda - 1) - 18] \Rightarrow (\lambda + 2) (\lambda^2 - 4\lambda - \lambda + 4 - 18)$$

$$\Rightarrow (\lambda + 2) (\lambda^2 - 5\lambda - 14) \Rightarrow (\lambda + 2) (\lambda^2 - 7\lambda + 2\lambda - 14)$$

$$(\lambda + 2) (\lambda + 2) (\lambda - 7)$$

$$\lambda = -2, \lambda = -2, \lambda = 7$$

$$\lambda = (7, -2, -2)$$

Now for B

~~$$\psi(\lambda) = |\lambda I - B|$$~~

$$\psi(\lambda) = |\mu I - B|$$

$$uI = \begin{vmatrix} u & 0 & 0 & 0 \\ 0 & u & 0 & 0 \\ 0 & 0 & u & 0 \\ 0 & 0 & 0 & u \end{vmatrix}, \quad uI - B = \begin{vmatrix} u-1 & -1 & -1 & -1 \\ -1 & u-1 & -1 & -1 \\ -1 & -1 & u-1 & -1 \\ -1 & -1 & -1 & u-1 \end{vmatrix}$$

$$|uI - B| = \begin{vmatrix} u-1 & -1 & -1 & -1 \\ -1 & u-1 & -1 & -1 \\ -1 & -1 & u-1 & -1 \\ -1 & -1 & -1 & u-1 \end{vmatrix}$$

$$R_1 + (R_2 + R_3 + R_4)$$

$$= \begin{vmatrix} u-4 & u-4 & u-4 & u-4 \\ -1 & u-1 & -1 & -1 \\ -1 & -1 & u-1 & -1 \\ -1 & -1 & -1 & u-1 \end{vmatrix}$$

Take $u-4$ common from R_1

$$= (u-4) \begin{vmatrix} 1 & 1 & 1 & 1 \\ -1 & u-1 & -1 & -1 \\ -1 & -1 & u-1 & -1 \\ -1 & -1 & -1 & u-1 \end{vmatrix}$$

$$R_2 + R_1, R_3 + R_1, R_4 + R_1$$

$$= (u-4) \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & u & 0 & 0 \\ 0 & 0 & u & 0 \\ 0 & 0 & 0 & u \end{vmatrix}$$

$$= (u-4) \cdot u^3 \quad \text{so } u=4, u=0, 0, 0$$

$$\mu = (4, 0, 0, 0)$$

Now $A \otimes B$

$\lambda \mu$

$$\lambda = (7, -2, -2), \mu = (4, 0, 0, 0)$$

$$\lambda \mu = (28, 0, 0, 0, -8, 0, 0, 0, -8, 0, 0, 0)$$